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WESAD Stress-detection

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Introducing WESAD, a Multimodal Dataset for Wearable Stress and Affect Detection

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ABSTRACT

Affect recognition aims to detect a person's affective state based on observables, with the goal to e.g. improve human-computer interaction. Long-term stress is known to have severe implications on wellbeing, which call for continuous and automated stress monitoring systems. However, the affective computing community lacks commonly used standard datasets for wearable stress detection which a) provide multimodal high-quality data, and b) include multiple affective states. Therefore, we introduce WESAD, a new publicly available dataset for wearable stress and affect detection. This multimodal dataset features physiological and motion data, recorded from both a wrist- and a chest-worn device, of 15 subjects during a lab study. The following sensor modalities are included: blood volume pulse, electrocardiogram, electrodermal activity, electromyogram, respiration, body temperature, and three-axis acceleration. Moreover, the dataset bridges the gap between previous lab studies on stress and emotions, by containing three different affective states (neutral, stress, amusement). In addition, self-reports of the subjects, which were obtained using several established questionnaires, are contained in the dataset. Furthermore, a benchmark is created on the dataset, using well-known features and standard machine learning methods. Considering the three-class classification problem (*baseline* vs. *stress* vs. *amusement*), we achieved classification accuracies of up to 80 %. In the binary case (*stress* vs. *non-stress*), accuracies of up to 93 % were reached. Finally, we provide a detailed analysis and comparison of the two device locations (*chest* vs. *wrist*) as well as the different sensor modalities.

KEYWORDS

Affective computing, Emotion recognition, Stress detection, Multimodal dataset, Sensor fusion, Benchmark, User study

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1 INTRODUCTION

Affective computing is an emerging field, inspired by the vision to improve human-computer interaction by building empathic machines. Empathic machines detect the affective state of a human user, adapt their 'behaviour' accordingly, and might even exhibit own emotional traits. From a health care point of view, stress, defined as 'nonspecific response of the body to any demand upon it' [25], is a particularly interesting affective state. This is due to the harmful effects of long-term stress, which can range from headaches and troubled sleeping to an increased risk of cardiovascular diseases [4, 16, 22]. According to the British Health and Safety Executive (HSE), stress accounted for 37% of all work-related ill health cases in 2015/16 [1]. These severe side effects of stress call for automated detection methods.

In order to build a reliable stress detection system, it is important to understand that stress is primarily a physiological response to a stimulus, triggered by the sympathetic nervous system (SNS). During this response a mixture of hormones like cortisol or adrenaline are released, leading to an increased breathing/heart rate and muscle tension. These physiological changes prepare the organism for a physical reaction ('fight-or-flight'). As shown by *Kreibitz et al.* [13] the physiological responses to certain emotional stimuli are also to some extent specific. A psychological model well suited for capturing affective states is Russel's circumplex model [23]. According to this model, affective states can be mapped into a 2D space, using for example the axes valence and arousal. The valence dimension indicates how negative/positive an affective state is perceived. On the arousal axis, which is known to be impacted by stress [9], the state is rated in terms of excitement.

In recent years, the specificity of the physiological responses to stress and emotional stimuli was utilised to train machine learning models to predict the affective state of a subject. Using deep neural networks, trained on audio and/or visual data, a high performance in emotion classification is achieved [17, 29]. However, these models are quite demanding in terms of computational resources and are only partially applicable on embedded devices. Classifying stress from audio samples was also successfully done, e.g., by Lu [14]. However, recording audio and/or video data continuously is in terms of privacy quite intrusive, and concerning technical feasibility difficult. Hence, these modalities are only available in specific occasions. Wearable electronic devices, in contrast, are only minimally intrusive. Devices like smart phones/watches are already popular among users. Contemporary wearables can be used to track steps and monitor other physical activities. To keep up with the current trend to quantify vital functions, a desirable next step is to infer affective states based on multimodal wearable sensor data.

WESAD: cosa contiene?

15
Soggetti usati

Chest
Sensore principale

3
Stati

Label considerate

1 = baseline → non-stress

2 = stress → stress

3 = amusement → divertito

Divisione dei notebook

Notebook 01:

- Esplorazione del dataset
- Caricamento di un file .pkl WESAD
- Ispezione del dizionario Python (chiavi chest/wrist/label)
- Visualizzazione dei segnali e analisi

Notebook 02:

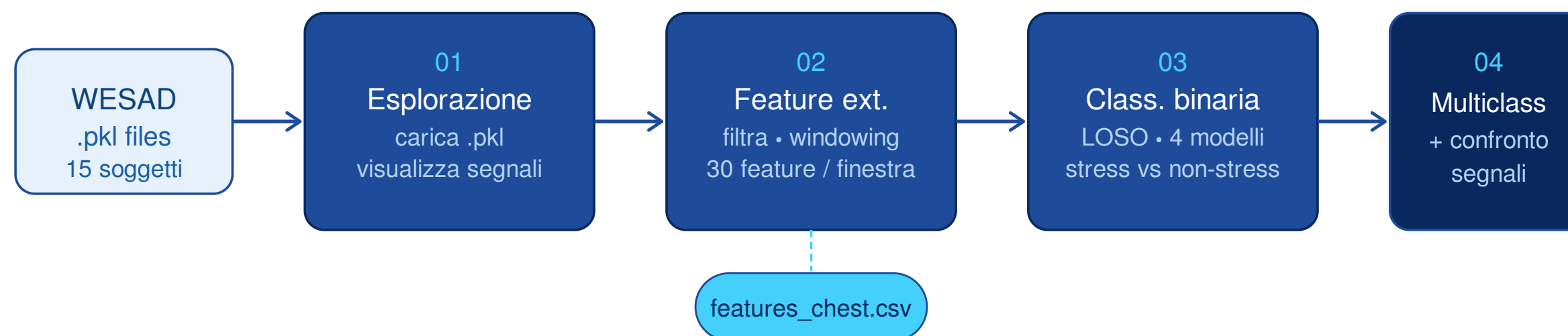
- Feature extraction
- Pre-processing
- Windowing dei segnali chest

Notebook 03:

- Classificazione binaria (stress vs non-stress)
- Training e confronto di 4 modelli
- Validazione Leave-One-Subject-Out
- Analisi delle metriche e della variabilità per ogni soggetto

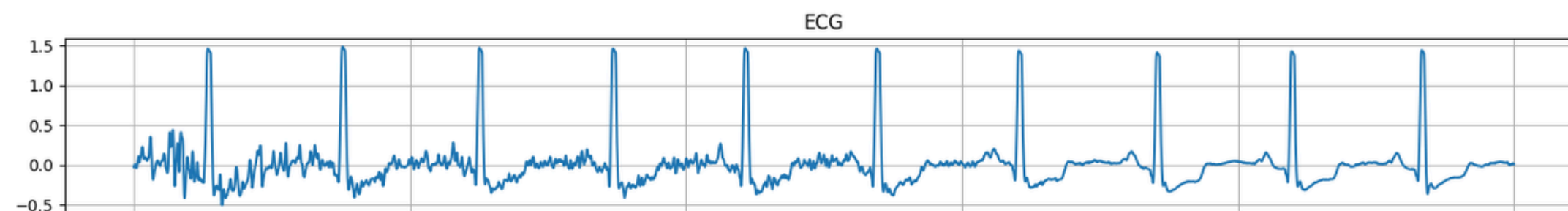
Notebook 04:

- Classificazione multiclasse (stress vs non-stress vs amusement)
- Confronto classificazione binaria vs multiclasse
- Confronto tra combinazioni di segnali

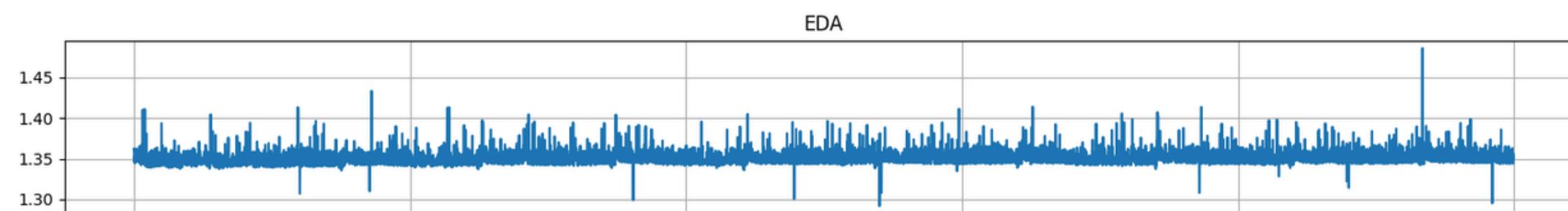


Segnali scelti

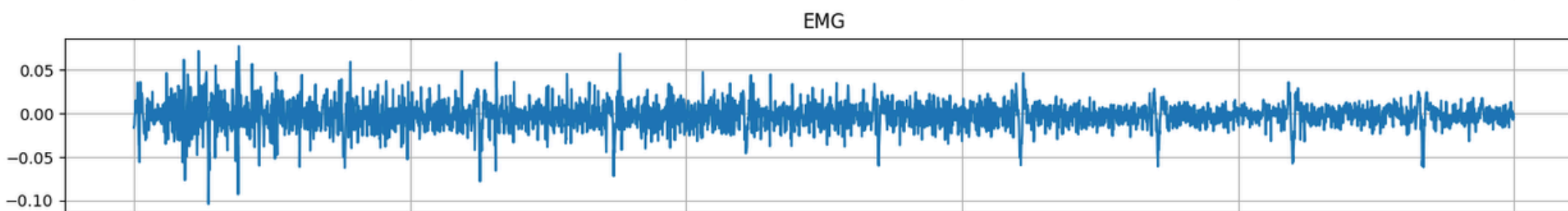
ECG



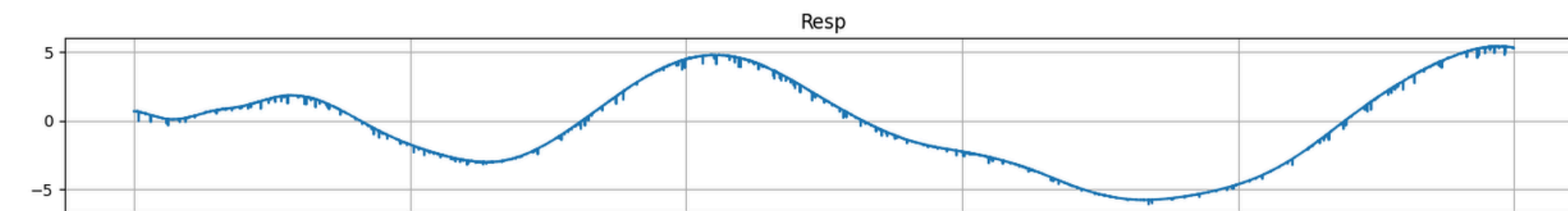
EDA



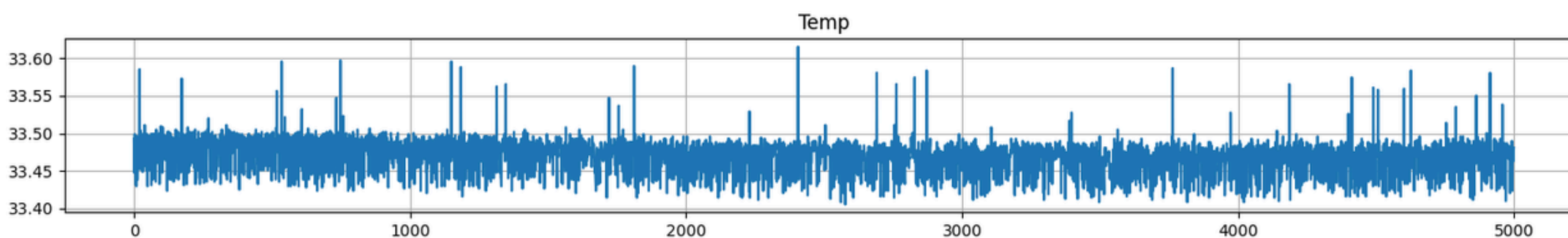
EMG



Resp



Temp



Notebook 02 – Feature extraction

I **segnali chest** sono segnali **grezzi** che non possono essere passati direttamente ai modelli.

Questo notebook li trasforma in un dataset tabellare.

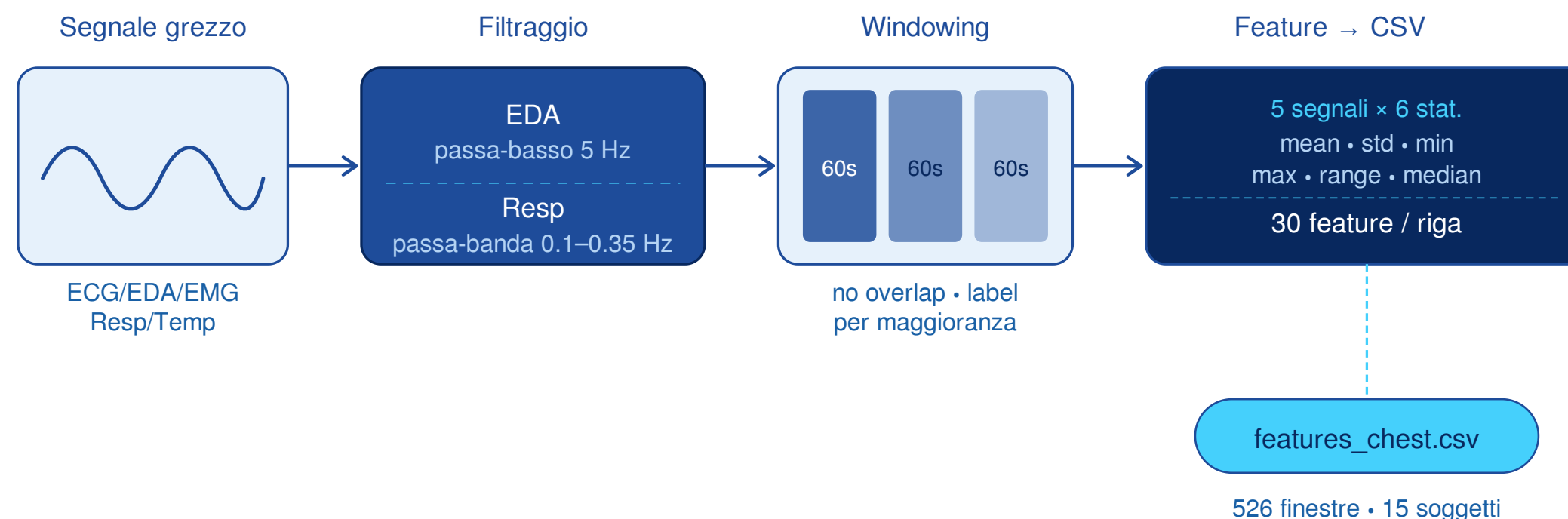
Pre-processing: filtro **passa-basso** (5 Hz) su EDA e **passa-banda** (0.1–0.35 Hz) su Resp, come indicato nel paper WESAD.

ECG, EMG e Temp lasciati **senza filtro**.

Windowing: i segnali vengono divisi in finestre da 60 secondi senza overlap. A ogni finestra viene assegnata la label prevalente (1, 2 o 3).

Feature: per ogni segnale si calcolano **6 statistiche** (media, std, min, max, range, mediana) → **30 feature totali** per finestra.

Output – features_chest.csv: 526 finestre su 15 soggetti (369 non-stress, 157 stress), con label originale, label binaria e ID soggetto.



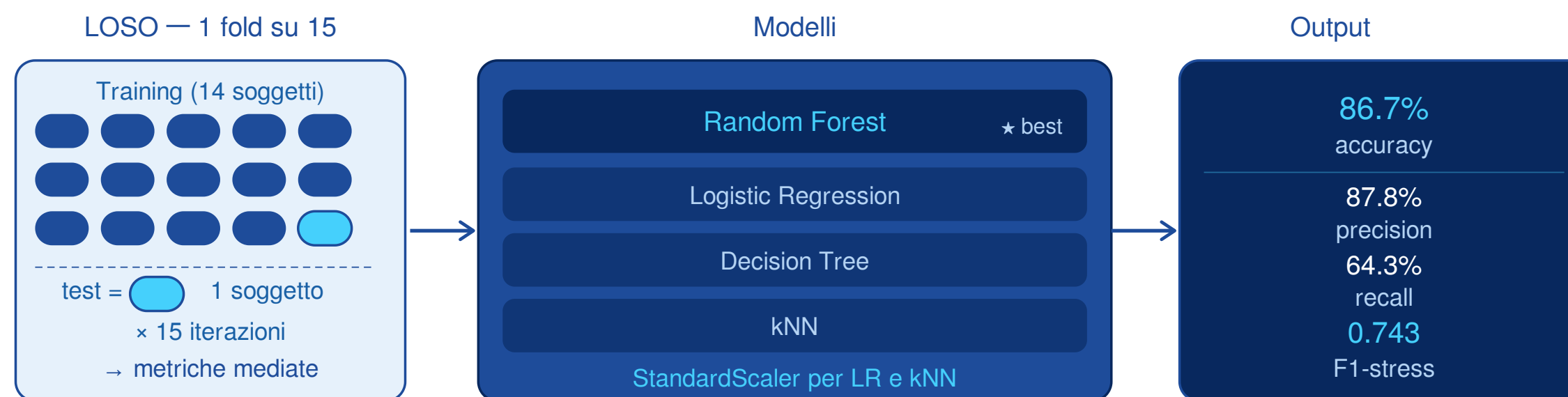
Notebook 03 – Classificazione binaria

Si carica `features_chest.csv` e si addestrano **4 classificatori** sul task stress vs non-stress, usando Leave-One-Subject-Out (**LOSO**) come schema di validazione: ogni iterazione lascia fuori **un soggetto** intero come test set, evitando che finestre dello stesso soggetto compaiano sia nel training che nel test.

Modelli: Logistic Regression, Decision Tree, Random Forest, kNN. Per LR e kNN si applica **StandardScaler**; DT e RF lavorano sulle **feature grezze**.

Risultati – Random Forest (miglior modello): **accuracy** 86.7%, **precision-stress** 87.8%, **recall-stress** 64.3%, **F1-stress** 0.743. La recall bassa indica che 56 finestre di stress vengono classificate come non-stress.

Analisi per soggetto: l'accuracy LOSO varia da 64.7% (S2, S10) a 100% (S6), evidenziando un'**alta variabilità** inter-soggetto nelle risposte fisiologiche.

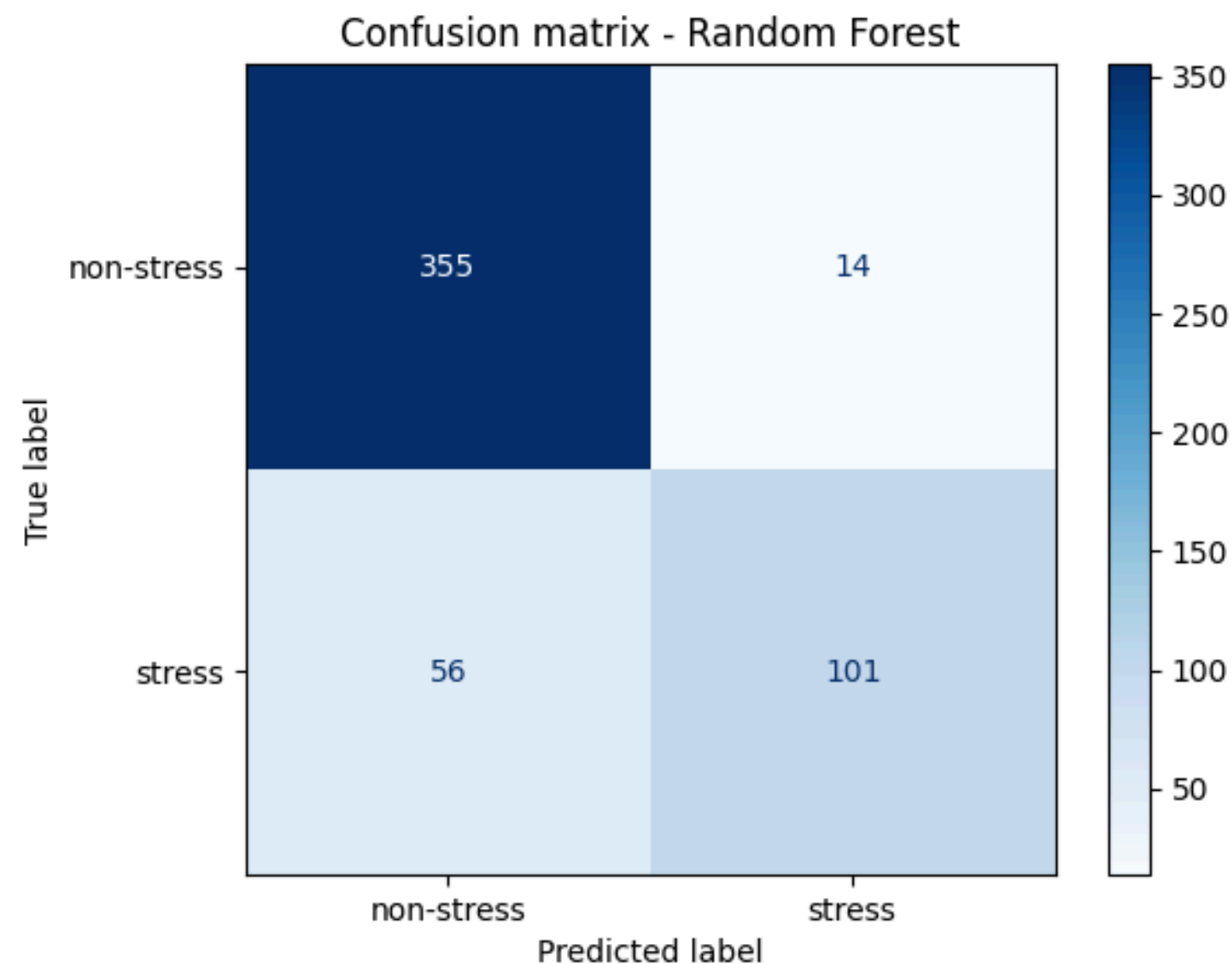


Classificazione Binaria

Modello	Accuracy	Precision	Recall	F1-Score
Random Forest	86.7%	87.8%	64.3%	74.3%
Decision Tree	80.8%	70.6%	61.1%	65.5%
kNN	81.9%	87.8%	45.9%	60.3%
Logistic Reg.	79.3%	62.4%	77.1%	68.9%

Random Forest ottiene l'F1 più alto sulla classe stress (0.743) con accuracy dell'86.7%. Logistic Regression ha recall più alta (0.771) ma precision inferiore.

Random Forest



86.7%
Accuracy

87.8%
Precision

64.3%
Recall

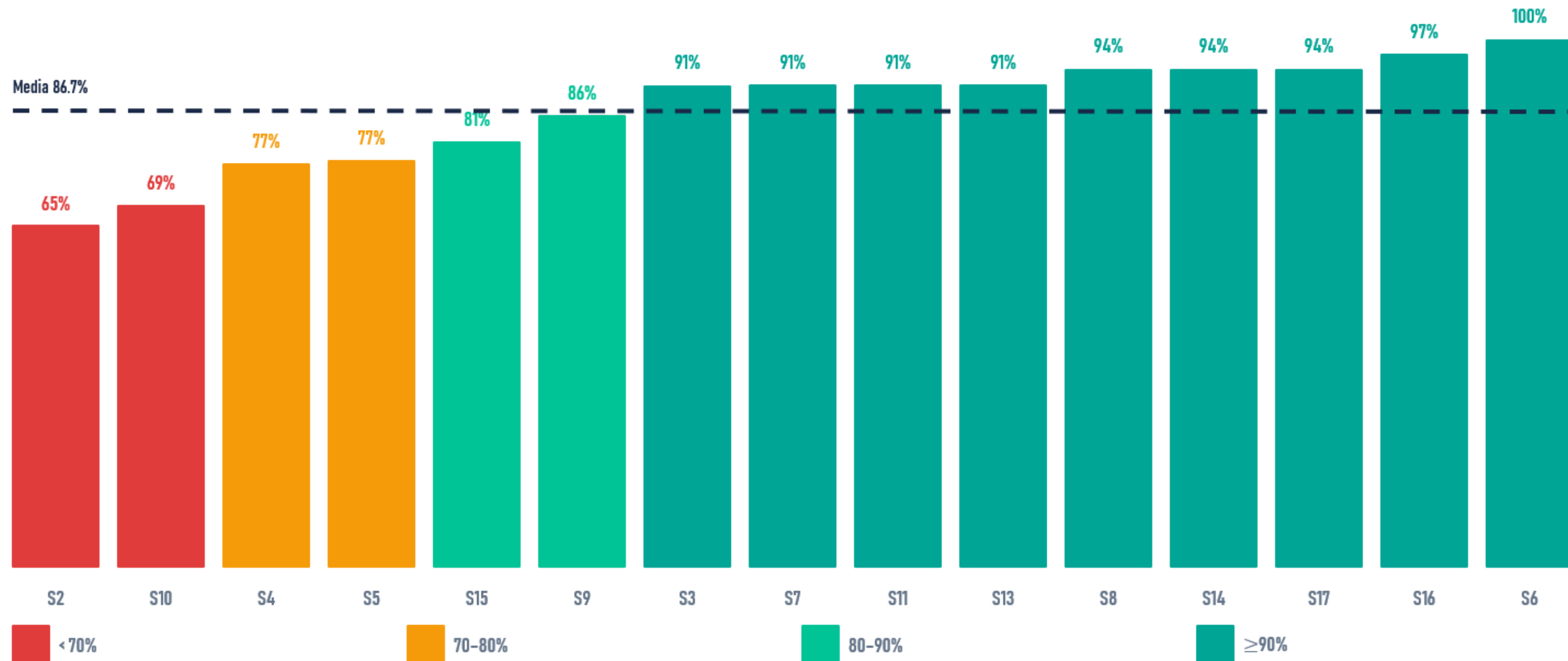
74.3%
F1-Score

La recall sullo stress è inferiore alla precision: 56 campioni di stress vengono erroneamente classificati come non-stress.



Variabilità per soggetto

Random Forest LOSO – Accuracy per ognuno dei 15 soggetti



Classificazione Multiclass

Modello	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Random Forest	73.6%	64.3%	61.1%	61.4%
Logistic Reg.	63.1%	58.5%	60.3%	58.8%
kNN	60.8%	55.3%	48.4%	49.9%
Decision Tree	60.1%	56.9%	52.4%	53.4%

Confronto Binary vs Multiclass – Random Forest

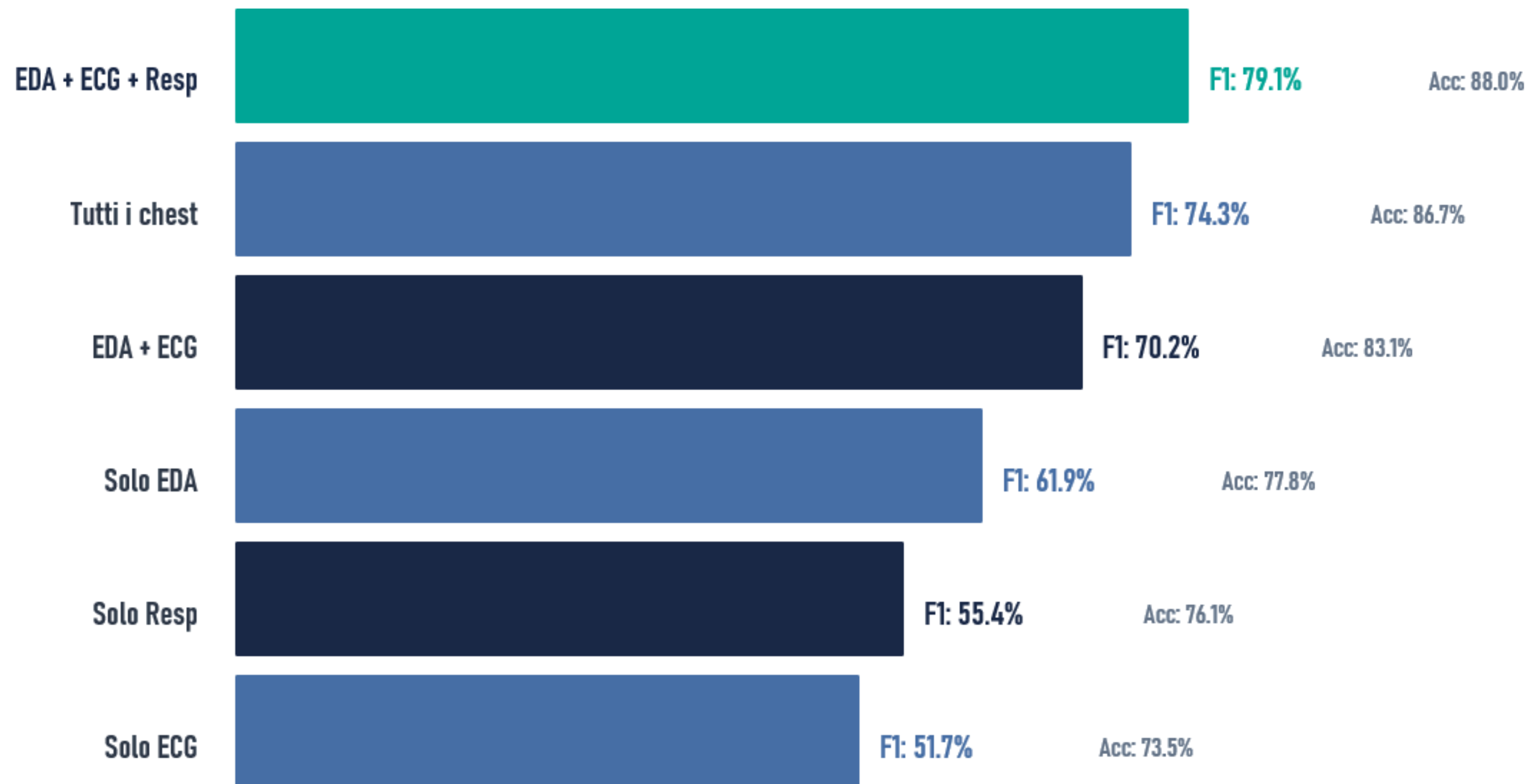
Binary
(Stress vs Non-Stress)

Accuracy: 86.7% · F1: 74.3%

Multiclass
(Base vs Stress vs Amus.)

Accuracy: 73.6% · F1: 61.4%

Confronto combinazioni di segnali



Confronto con il Paper Originale

	Paper WESAD (2018)	Nostro Progetto
Segnali	Chest + Wrist (tutti)	Solo chest (ECG, EDA, EMG, Resp, Temp)
Feature	Fisiologiche specializzate ~100	Statistiche semplici ~34
Finestre	60s, overlap 0.25s	60s senza overlap
Modelli	DT, RF, AB, LDA, kNN	LR, DT, RF, kNN
Binario migliore	LDA: 93.1%	RF: 86.7% (acc), F1=74.3%
3 classi migliore	AB: 80.3%	RF: 73.6% (acc), F1=61.4%
Validazione	LOSO	LOSO

Conclusioni

I segnali fisiologici

EDA e Resp sono i più informativi.
Con feature semplici, RF raggiunge ~87% di accuracy.

La qualità delle feature conta

Feature fisiologiche specializzate (come nel paper) migliorano significativamente i risultati.

Alta variabilità tra i soggetti

Da 64.7% a 100% di accuracy: le risposte fisiologiche sono individuali.
Servono modelli personalizzati.

Task binario vs multiclass

Stress/non-stress è più semplice del 3-class:
baseline e amusement sono fisiologicamente simili.

Meno feature ≠ risultati peggiori

EDA+ECG+Resp ha un esito migliore rispetto a tutti i segnali:
selezionare solo i segnali migliori riduce il rumore.

LOSO è la valutazione migliore

Leave-One-Subject-Out stima realisticamente la generalizzazione su soggetti non visti.